

VITAMIN B12

Methylcobalamin — Complete Injection Guide — 10mg Vials

Active Form of Cobalamin | Bioavailable | No Conversion Required

Purity 99.8% | Endotoxin-Free | Heavy Metal-Free

PRODUCT SPECIFICATIONS

Parameter	Detail
Product	Vitamin B12 — Methylcobalamin
Form	Methylcobalamin (active, bioavailable cobalamin)
Vial Content	10 mg lyophilized powder per vial
Quantity	10 vials (100 mg total)
Purity	99.8%
Endotoxin	< 0.1 EU/mg
Heavy Metals	Below detectable limits
Reconstitution	1 mL BAC Water → 10 mg/mL
Routes	Intramuscular (IM) preferred, or Subcutaneous (SubQ)
Storage (powder)	2-8°C refrigerated, protect from light
Storage (recon.)	2-8°C, use within 28 days, protect from light
Appearance (recon.)	Pink to dark red solution — normal for B12

WHAT IS VITAMIN B12 (METHYLCOBALAMIN)?

Vitamin B12 (cobalamin) is an essential water-soluble vitamin required for red blood cell formation, neurological function, DNA synthesis, and cellular energy metabolism. Methylcobalamin is the active, bioavailable form of B12 — unlike cyanocobalamin, it requires no hepatic conversion and is immediately available for enzymatic reactions in the body.

Mechanism of Action: Methylcobalamin serves as a cofactor for methionine synthase (converting homocysteine to methionine) and for mitochondrial energy production via the methylmalonyl-CoA mutase pathway. It is critical for myelin sheath maintenance, neurotransmitter synthesis, and proper cell division. Injectable B12 bypasses gastrointestinal absorption barriers, achieving far higher serum levels than oral

supplementation.

Why Injectable? Many individuals have impaired B12 absorption due to low intrinsic factor, gastrointestinal conditions (Crohn's, celiac, gastric bypass), medications (metformin, PPIs), or age-related decline. Intramuscular or subcutaneous injection delivers B12 directly into the bloodstream, bypassing all absorption barriers for maximum bioavailability.

KEY RESEARCH BENEFITS

Benefit	Description
Energy & Metabolism	Essential cofactor in mitochondrial ATP production; reduces fatigue and supports cellular energy metabolism
Neurological Health	Maintains myelin sheath integrity; supports nerve regeneration and cognitive function
Red Blood Cell Formation	Required for proper erythropoiesis; prevents megaloblastic anemia and associated symptoms
Homocysteine Regulation	Converts homocysteine to methionine, reducing cardiovascular risk associated with elevated homocysteine
DNA Synthesis	Cofactor in nucleotide synthesis; essential for cell division, tissue repair, and immune cell production
Mood & Sleep	Supports serotonin and melatonin synthesis; may improve mood, focus, and circadian rhythm regulation
Methylation Support	Donates methyl groups for epigenetic regulation, detoxification, and neurotransmitter production

RECONSTITUTION PROTOCOL



Temp 20°C

Both BAC Water and vial



Wipe caps

Alcohol 70% both stoppers



Draw water

Exact volume into syringe



Inject wall

Along glass wall — slow



Verify

Pink/red solution normal color

Step	Action	Details
1	Temperature equilibrium	Allow BAC Water and B12 vial to reach room temperature (20°C) before mixing
2	Sterilize	Wipe the rubber stopper of both the BAC Water vial and B12 vial with 70% isopropyl alcohol swab
3	Draw BAC Water	Using a sterile syringe, draw exactly 1.0 mL of Bacteriostatic Water

4	Inject along wall	Insert needle through B12 vial stopper. Slowly release BAC Water along the inside glass wall — do NOT spray directly onto the powder
5	Dissolve gently	Swirl the vial gently until fully dissolved. Solution will be PINK to DARK RED — this is NORMAL for methylcobalamin. Never shake vigorously.

■■ **COLOR NOTE:** Reconstituted methylcobalamin produces a distinctive PINK to DARK RED solution. This is completely normal and indicates the cobalt-containing cobalamin ring structure. A colorless solution would indicate degradation — discard if colorless.

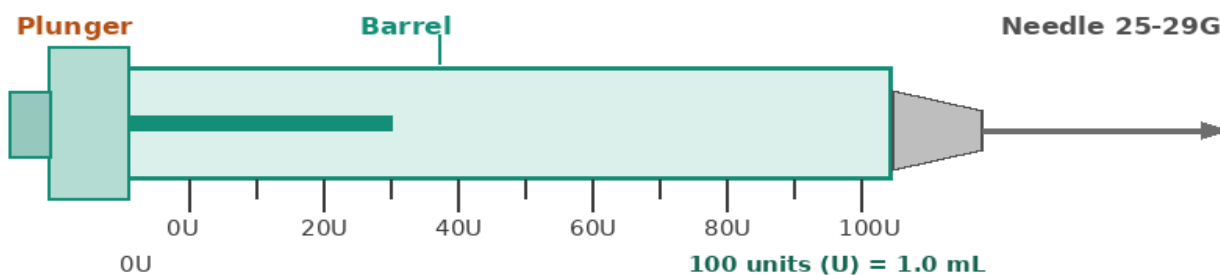
Vial Size	BAC Water	Concentration	1 mg Dose	2.5 mg Dose	5 mg Dose
10 mg	1.0 mL	10 mg/mL	10 units	25 units	50 units

DOSE CALCULATION

Formula: $\text{Dose (units)} = \text{Desired mg} \div \text{Concentration (mg/mL)} \times 100$

10mg Vial reconstituted with 1 mL BAC Water → 10 mg/mL:

Desired Dose	Calculation	Draw Volume	Doses per Vial
1.0 mg (1000 mcg)	$1.0 \div 10 \times 100 = 10 \text{ units}$	10 units (0.10 mL)	10 doses
2.5 mg (2500 mcg)	$2.5 \div 10 \times 100 = 25 \text{ units}$	25 units (0.25 mL)	4 doses
5.0 mg (5000 mcg)	$5.0 \div 10 \times 100 = 50 \text{ units}$	50 units (0.50 mL)	2 doses
10.0 mg (full vial)	$10.0 \div 10 \times 100 = 100 \text{ units}$	100 units (1.00 mL)	1 dose

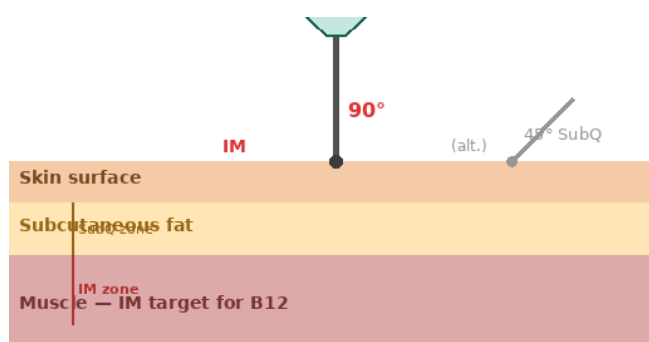
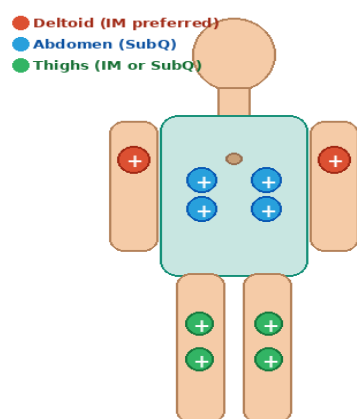


INJECTION TECHNIQUE

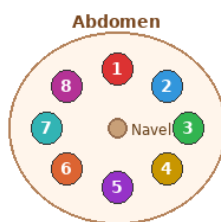
Preferred Route: Intramuscular (IM) — B12 has superior absorption and longer duration when injected into muscle tissue. Subcutaneous (SubQ) is an acceptable alternative for those who prefer it.

Step	Action	Details
1	Wash hands	Wash hands thoroughly with soap and water. Use clean exam gloves if available.

2	Prepare syringe	Draw the calculated dose from the reconstituted vial using a sterile syringe. For IM: use 25-27G, 1-inch needle. For SubQ: use 29-31G insulin syringe.
3	Remove air	Hold syringe needle-up, tap barrel gently, push plunger until a tiny drop appears at needle tip.
4	Clean site	Wipe the injection site with a 70% isopropyl alcohol swab and allow to air dry for 30 seconds.
5	IM technique	For IM: Stretch skin taut over deltoid or outer thigh. Insert needle at 90° angle with a firm, swift motion.
6	SubQ technique	For SubQ: Pinch a fold of skin at the abdomen or thigh. Insert needle at 45° angle into the skin fold.
7	Inject & withdraw	Depress the plunger slowly and steadily over 5-10 seconds. Withdraw needle, apply gentle pressure with a cotton ball.



Site Rotation Map — Rotate injection sites to prevent tissue irritation:



Rotate: 1->2->...->8->1

DOSING PROTOCOL & SCHEDULE

Injection days (3x/week example)



Alternate: 1x weekly or 2x weekly depending on protocol
Inject any time of day — no fasting required

Protocol	Dose	Frequency	Duration
Loading (deficiency)	5-10 mg	3x per week (IM)	4 weeks
Standard maintenance	1-2.5 mg	1-2x per week (IM/SubQ)	Ongoing
Enhanced energy	2.5-5 mg	2-3x per week	8-12 weeks, then reassess
Neurological support	5 mg	3x per week (IM)	8-12 weeks, then reduce
General wellness	1 mg	1x per week	Ongoing maintenance

Phase	Duration	Notes
Loading Phase	4 weeks	Higher doses 3x/week to rapidly replenish depleted stores. Essential for diagnosed deficiency.
Transition Phase	4 weeks	Reduce frequency to 2x/week. Monitor energy levels and symptoms.
Maintenance Phase	Ongoing	1x weekly injection at 1-2.5 mg. Adjust based on serum B12 levels and symptom response.
Reassessment	Every 3-6 months	Check serum B12 and methylmalonic acid levels. Adjust protocol based on lab results.

Note: B12 is water-soluble — excess is excreted via the kidneys. There is no established upper toxicity limit for B12 injection, and it is generally considered very safe even at high doses. However, always follow your healthcare provider's recommended protocol.

SIDE EFFECTS & SAFETY

Side Effect	Frequency	Management
Injection site soreness	Common	Rotate sites. Apply gentle pressure post-injection. Resolves within 1-2 days.
Pink/red urine	Common	Normal — excess B12 is excreted renally. Not a cause for concern.
Mild warmth/flushing	Uncommon	Transient vasodilation effect. Resolves within 30 minutes.
Mild headache	Uncommon	Stay hydrated. Usually resolves within hours. Reduce dose if persistent.

Mild acne breakout	Uncommon	May occur during loading phase due to metabolic upregulation. Usually temporary.
Diarrhea	Rare	Reduce dose or frequency. Ensure adequate hydration.

■■ **STOP USE AND CONSULT A PHYSICIAN IF:**

- Severe allergic reaction (hives, swelling, difficulty breathing, anaphylaxis)
- Chest pain, rapid heartbeat, or shortness of breath
- Severe or persistent headache, blurred vision
- Signs of infection at injection site (increasing redness, warmth, pus, fever)
- Any unexpected or concerning symptoms

Contraindications: Individuals with a known allergy to cobalamin or cobalt should not use this product. Those with Leber's hereditary optic neuropathy should avoid B12 supplementation. Patients on certain medications (chloramphenicol, colchicine, aminosalicic acid) should consult their physician as these may interact with B12 metabolism.

STORAGE GUIDE

State	Temperature	Duration	Notes
Lyophilized powder (sealed)	2-8°C refrigerated	Until expiry date	Keep in original packaging; PROTECT FROM LIGHT
Reconstituted solution	2-8°C refrigerated	28 days maximum	Label with reconstitution date. Protect from light — wrap in foil.
BAC Water (opened)	2-8°C refrigerated	28 days	Use dedicated BAC Water vial. Do not contaminate.
During transport	Cool pack / insulated	Minimize time	Avoid temperatures above 25°C. Protect from direct sunlight.

■ **LIGHT SENSITIVE:** Methylcobalamin degrades when exposed to light. Store reconstituted vials wrapped in aluminum foil or in an opaque container. Avoid leaving vials in direct sunlight or under fluorescent lighting for extended periods. Solution should remain pink/red — discard if it becomes colorless.

GOLDEN RULES

#	Rule
1	Always use a NEW sterile syringe and needle for each injection — never reuse.
2	Reconstitute with Bacteriostatic Water (0.9% benzyl alcohol). Never use saline or plain sterile water.
3	Inject BAC Water slowly along the glass wall — never spray directly onto the powder.

4	Store reconstituted vials at 2-8°C, PROTECTED FROM LIGHT, and use within 28 days.
5	The reconstituted solution is PINK to DARK RED — this is normal. Discard if colorless.
6	For IM injection, use a 25-27G, 1-inch needle at 90° into the deltoid or outer thigh.
7	For SubQ injection, use a 29-31G insulin syringe at 45° with a skin pinch.
8	Rotate injection sites systematically to prevent tissue irritation and soreness.
9	Clean both the injection site AND vial stopper with 70% alcohol before every injection.
10	Pink/red urine after injection is NORMAL — excess B12 is safely excreted by the kidneys.

DISCLAIMER: This guide is provided for research and educational purposes only. This product is sold as a research compound and is not intended to diagnose, treat, cure, or prevent any disease. Consult a qualified healthcare professional before using any injectable supplement. SaiyanMed assumes no liability for misuse of this product or information.

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